Frequency Forensics and Digital Detective

Name: AASHISH

Roll Number: 240021

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Abstract

This project investigates two important image processing applications. The first task focuses on frequency-domain image restoration using the Fourier Transform to recover information hidden by periodic interference. The second task focuses on edge detection using the Sobel operator to identify object boundaries within an image. Both tasks demonstrate how signal processing techniques can be applied to image analysis, restoration, and feature extraction.

Q1A: Frequency Forensics – "The Ghost Signal"

Objective

The objective of this task is to recover an image corrupted by periodic interference using frequency-domain filtering techniques. The image is transformed into the frequency domain using the 2D Discrete Fourier Transform (DFT), noise frequencies are identified and suppressed, and the image is reconstructed using the inverse transform.

Theory

A two-dimensional image can be represented as a discrete signal I(x,y). The 2D Discrete Fourier Transform is given by:

$$F(u,v) = \overset{\Sigma^{I}}{\sum} \quad ▒x, y)e^{-j2\pi((ux/M)+(vy/N))}$$

where M and N represent the image dimensions. The inverse transform is defined as:

$$I(x,y) = (1/MN) SumSum F(u,v)e^{(j2\pi[(ux/M)+(vy/N)])}$$

Periodic interference appears as isolated, bright peaks in the frequency spectrum. By removing these peaks using a carefully designed notch filter, the unwanted frequencies can be suppressed while preserving useful structural image information.

Methodology & Parameter Justification

1. **Transformation:** Load the corrupted image and compute the 2D FFT. Shift the spectrum to the center to standardize the coordinate system (low frequencies at the core, high frequencies at the edges).
2. **Analysis:** Compute the magnitude spectrum in linear and dB (logarithmic) scales. The dB scale is necessary to visualize and identify the hidden periodic noise frequencies.
3. **Filter Design (Justification):** A custom notch reject filter was constructed. A dynamic **99.8th percentile intensity threshold** was utilized to target and isolate only the extreme, anomalous noise spikes without accidentally deleting standard high-frequency edge data. Additionally, a **25-pixel protection radius** was enforced at the center of the spectrum. This is crucial because the central core contains the low-frequency components representing the fundamental shapes and text; protecting it ensures the recovered image does not become severely blurred.
4. **Reconstruction:** Apply the filter mask in the frequency domain, perform the inverse FFT, and reconstruct the recovered spatial image.

Results

(Insert the following figures from your Colab output)

- **Figure 1:** Corrupted Input Image
- **Figure 2:** Magnitude Spectrum (Linear Scale)
- **Figure 3:** Magnitude Spectrum (dB Scale)
- **Figure 4:** Applied Notch/Peak Mask
- **Figure 5:** Filtered Spectrum (dB Scale)
- **Figure 6:** Recovered Image

Observations & Discussion

The Fourier spectrum initially showed concentrated energy around the center, with the periodic interference appearing as highly distinct, isolated peaks extending away from the origin (Figure 3). The notch filter successfully targeted and zeroed out these unwanted frequency spikes without damaging the central core (Figures 4 and 5).

Conclusion

The frequency-domain image recovery system successfully eliminated the periodic interference. The experiment proves the effectiveness of Fourier-based filtering, successfully restoring the hidden image and clearly revealing the previously obscured text: *"QUIZ 2 ON 7TH JULY IN TUTORIAL HOURS."*

Q1B: Digital Detective – "Missing Boundaries"

Objective

The objective of this task is to detect object boundaries using the Sobel edge detection operator. Edge information is extracted by computing image gradients in horizontal and vertical directions while mitigating background noise.

Theory

Edges correspond to regions where image intensity changes rapidly. The continuous image gradient is defined as:

$$\nabla I = [\partial I/\partial x, \partial I/\partial y]$$

The overall gradient magnitude, which determines edge strength, is:

$$|\nabla I| = \sqrt{G_x^2 + G_y^2}$$

where G_x is the horizontal gradient and G_y is the vertical gradient. For discrete digital images, the Sobel operator approximates these continuous derivatives using 3×3 convolution kernels:

Horizontal Sobel Kernel

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Vertical Sobel Kernel

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Methodology & Parameter Justification

1. **Pre-processing (Justification):** The image is loaded in grayscale. Before edge extraction, a **5×5 Gaussian blur** is applied. Because numerical differentiation is highly sensitive to high-frequency variations, random pixel noise can easily manifest as false, sharp edges. The Gaussian blur smooths the image, suppressing background noise so the Sobel filter only triggers on valid, physical structural boundaries.

2. **Extraction:** Apply the Sobel-X operator to detect vertical boundaries and the Sobel-Y operator to detect horizontal boundaries.
3. **Combination:** Compute the Euclidean gradient magnitude to merge both directional maps into a single, comprehensive edge representation.

Results

(Insert the following figures from your Colab output)

- **Figure 7:** Original Input Image
- **Figure 8:** Smoothed Image (Noise Filtered)
- **Figure 9:** Sobel X (Vertical Boundaries)
- **Figure 10:** Sobel Y (Horizontal Boundaries)
- **Figure 11:** Combined Gradient Magnitude

Observations & Discussion

Strong edges were successfully detected around the pinwheel boundaries and the central hub produced high gradient values. As expected, the Sobel-X operator successfully isolated the vertical spokes, while the Sobel-Y operator highlighted the horizontal spokes. The application of the 5×5 Gaussian smoothing visibly improved edge continuity and prevented the background texture from creating false-positive edge signals.

Conclusion

The Sobel edge detector successfully extracted the structural boundaries of the objects present in the image. The experiment demonstrates the robustness of gradient-based techniques for feature extraction, provided that adequate low-pass filtering (smoothing) is applied beforehand to mitigate noise.

Overall Conclusion

The project successfully demonstrated two foundational image processing applications. Frequency-domain filtering (via DFT) is a powerful tool for recovering images corrupted by periodic interference, while spatial gradient-based edge detection (via Sobel) is highly effective for identifying object boundaries. These techniques form the backbone of broader applications in communication systems, surveillance, medical imaging, and autonomous computer vision.